VOLTAGE ASSIGNMENTS ON A 44-VERTEX RECONSTRUCTION OF H_{12}: GIRTH 12 FOR $m = 7, 8, 9$, AND NONE FOR $3 \leq m \leq 6$

Abstract. Aguilar, Araujo-Pardo and Berman exhibit a voltage graph H_{12} on 44 vertices whose $\mathbb{Z}_m$ lifts are $(\{3, m\}; 12)$-graphs of order $41m+3$ for every $m \geq 10$, and ask, in the third sub-bullet under "Related to semicubic graphs" of their Section 6 (p. 29), for voltage assignments on that same H_{12} producing girth 12 for $m \in \{3, 4, \ldots, 9\}$, or a proof that none exists. This note is about one explicit 44-vertex voltage graph, printed in full in Section 2, which is our reading of the source's Figure 16 and its Tables 3–4; we do *not* verify that it is the source's H_{12}, and we therefore do not claim to settle the cited question — whether we do turns entirely on that identification, which is stated and bounded in Section 7. For the graph printed below: girth-12 assignments exist for $m = 7, 8, 9$ — exhibited here in full, at orders 290, 331 and 372 — and none exists for $m = 3, 4, 5, 6$. The two smaller witnesses are checked as graphs in their own right, so they give

$$n(\{3, 7\}; 12) \leq 290 \ \text{(was 334)}, \qquad n(\{3, 8\}; 12) \leq 331 \ \text{(was 374)},$$

independently of the identification; both lie strictly above the published lower bounds 272 and 308. The $m = 9$ witness carries *no* record claim: Goedgebeur, Jooken and Van den Eede publish $n(\{3, 9\}; 12) \leq 360 < 372$. The second question of the same sub-bullet, which asks for a *different* graph H'_{12}, is untouched and remains open at every m.

1. The question

Following Aguilar, Araujo-Pardo and Berman [1], a $(\{3, m\}; g)$-*graph* is a graph whose every vertex has degree 3 or m and whose girth is g; $n(\{3, m\}; g)$ denotes the least order of such a graph. In a voltage graph over $\mathbb{Z}_m$ a *pinned* vertex v^* is a base vertex of degree 1 whose lift is a *single* vertex joined to all m copies of its unique neighbour, so that $\deg v^* = m$; a base whose other vertices have degree 3 and which carries s pinned vertices therefore lifts to a graph with exactly s vertices of degree m and all others of degree 3 [1, Remark 3.1]. Unlabelled edges carry voltage 0.

The source's Theorem 5.6 (p. 27) reads, verbatim:

> "The graph $(H_{12}; m)$ given by the voltage graph H_{12} shown in Figure 16 with voltages given in Table 4 has girth 12 for $m \geq 10$. Moreover, it is a $(\{3, m\}; 12)$-graph with three vertices of degree m and $41m$ vertices of degree 3."

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Key words and phrases. bi-regular cage, semicubic graph, girth, voltage graph, lift, generalized hexagon, exhaustive search.

Our target is the *first sentence* of the third sub-bullet under the bullet "Related to semicubic graphs" in Section 6, "Open questions", of [1], on p. 29, verbatim (subscripts written H_{12}, T_{12}, H'_{12}):

> "Find different voltage assignments for the graph H_{12} that produce graphs of girth 12 for $m \in \{3, 4, \ldots, 9\}$, or show no assignments exist. *Preliminary investigations suggest that new arc assignments are required to find smaller examples.* Does there exist a different graph H'_{12} (for example, using the same modified tree structure and arcs a, b, c, d but different arcs among the remaining leaves of T_{12}) that produces graphs of girth 12 for these smaller values of m? For example, can we find graphs of small girth if we construct a 16-cycle among the remaining leaves of T_{12}, or if we interlace two 8-cycles differently?"

The parent bullet, pp. 28–29, reads "Related to semicubic graphs: Find voltage assignments to construct graphs of girth 10 and 12 for missing values of m. More specifically:". The locator for the target is therefore: [1], Section 6 "Open questions", third sub-bullet under "Related to semicubic graphs", p. 29.

Scope. That sub-bullet contains *two* questions. Everything below concerns the first, and only in the form it takes for the explicit graph of Section 2; nothing here bears on the second, the different graph H'_{12}. Nothing here may be read as "the third open question is closed": that reading needs the identification of Section 7, which is not established here.

Why $m = 9$ was singled out by the authors, [1, p. 28], verbatim:

> "To compare the parameters we recall that the lower bound given in Lemma 3.4 in [6] is $n(\{3, m\}; 12) \geq \lceil 109m/3 \rceil + 17$. The graphs (G_{12}, m) with $m \geq 9$ has order $49m + 3$ whereas the graphs (H_{12}, m) with $m \geq 10$ has order $41m + 3$. . . . Interestingly, if we are able to find a voltage assignment to construct an $(H_{12}, 9)$ we would be able to construct a graph of order 372."

That order-372 graph is exhibited in Section 4. The lower bound quoted there is attributed by [1] to Lemma 3.4 of Araujo-Pardo, Balbuena, García-Vázquez, Marcote and Valenzuela [4].

2. The 44-vertex base graph, in full

Everything below is a statement about one explicit 44-vertex graph, so that graph is printed here completely: a reader needs neither the source's figure nor any program. It is our reading of the graph of [1, Figure 16]; that reading, and the two places where the source's printed tables have to be read against their own deletion list to obtain it, are recorded in Section 7, which also states what has *not* been verified about the identification. *Convention.* From here on "H_{12}" abbreviates the graph printed in this section, and nothing else; it is not asserted to be the graph of [1, Figure 16].

The 44 **vertices** (x^*, y^*, z^* are the three pinned vertices):

```
x*, x, x0, x1, x00, x01, x10, x11, x010, x011, x100, x101, x110,
x111, x0100, x0101, x0110, x0111, x1010, x1011, x1110, x1111,
y*, y, y0, y1, y01, y10, y11, y010, y011, y101, y110,
z*, z, z0, z1, z01, z10, z11, z010, z011, z101, z110
```

The 43 **tree edges, all at voltage** 0**:**

```
x*-x, x-x0, x-x1, x0-x00, x0-x01, x1-x10, x1-x11, x01-x010, x01-x011,
x10-x100, x10-x101, x11-x110, x11-x111, x010-x0100, x010-x0101,
x011-x0110, x011-x0111, x101-x1010, x101-x1011, x111-x1110, x111-x1111,
y*-y, y-y0, y-y1, y0-y01, y1-y10, y1-y11, y01-y010, y01-y011,
y10-y101, y11-y110,
z*-z, z-z0, z-z1, z0-z01, z1-z10, z1-z11, z01-z010, z01-z011,
z10-z101, z11-z110,
x00-y0, x00-z0
```

The 20 **labelled arcs** (name: tail $\to$ head). These are the only edges carrying a voltage, and they are the 20 coordinates of every voltage tuple in this note, always in this order:

```
a: x100 ->y10      b: x100 ->z10      c: x110 ->y11      d: x110 ->z11
e: x0100->y010     j: x0101->y011     q: x0110->y101     u: x0111->y110
w: x1010->y110     s: x1011->y101     l: x1110->y011     g: x1111->y010
f: x0100->z101     k: x0101->z110     r: x0110->z010     v: x0111->z011
al:x1010->z010     t: x1011->z011     p: x1110->z101     h: x1111->z110
```

(the arc written `al` is the source's α). The coordinate order is

$$\varphi = [\, a, b, c, d, e, j, q, u, w, s, l, g, f, k, r, v, \alpha, t, p, h \,].$$

The lift. For φ: {arcs} $\to \mathbb{Z}_m$, the derived graph (H_{12}, φ) has, for each non-pinned base vertex v, the m vertices (v, i), $i \in \mathbb{Z}_m$, together with the three single vertices x^*, y^*, z^*. A tree edge u–v contributes (u, i)–(v, i) for every i; an arc $X : t \to h$ contributes (t, i)–$(h,\, i + \varphi(X) \bmod m)$ for every i; and x^* is joined to (x, i) for all i, likewise y^* to (y, i) and z^* to (z, i).

Proposition 1. *The graph above has* 44 *vertices and* 63 *edges, no loop and no repeated edge; its degree sequence is* 1 *three times and* 3 *forty-one times, and the three vertices of degree* 1 *are exactly* x^*, y^*, z^**; the* 43 *tree edges form a spanning tree, so the cycle rank is* $63 - 44 + 1 = 20$ *and the* 20 *arcs are a cotree; and it is bipartite, with classes of sizes* 23 *and* 21*, every one of the* 20 *arcs joining opposite classes. Consequently* (H_{12}, φ) *has order* $41m + 3$ *and* $63m$ *edges, exactly three vertices of degree* m *and* $41m$ *of degree* 3*, and is bipartite.*

Restoring the eight leaves that [1, p. 27] deletes from T_{12} to form H_{12} — namely $x1000$, $x1001$, $x1100$, $x1101$, $y100$, $y111$, $z100$, $z111$, attached to $x100, x100, x110, x110, y10, y11, z10, z11$ respectively — gives a tree on 52 vertices with 51 edges, hence 49 non-pinned vertices. That reproduces the source's own $49m + 3$ and its "$441 = 49 \cdot 9$ vertices of degree 3" for the companion family G_{12}, and $52-8 = 44$, $51-8 = 43$ reproduce Theorem 5.6's "$41m$ vertices of degree 3".

3. The search space, and the necessary condition

Why $\mathbb{Z}_m^{20}$ with the tree at 0 is the whole space. Two independent reasons. First, it is literally the space the question asks about: in [1] the graph H_{12} carries voltages on exactly the 20 labelled arcs of Table 4, and every other edge is unlabelled, i.e. at voltage 0. Second, it is gauge-complete anyway. *Switching* — re-indexing the fibre over each base vertex by an arbitrary $f : V \to \mathbb{Z}_m$ — zeroes the voltages on any spanning tree without changing the derived graph up to isomorphism. By Proposition 1 the 43 tree edges *are* a spanning tree and the cycle rank is 20, so the normalised space has dimension exactly the cycle rank. The pinned vertices do not break this: a pinned vertex is joined to *all* m copies of its unique neighbour, so the voltage on its pendant edge is irrelevant, and switching at that neighbour merely permutes the fibre cyclically, which the pinned join is invariant under. A census of $\mathbb{Z}_m^{20}$ therefore meets every isomorphism class of lift and decides the cell as posed.

The short-cycle system. A base cycle of length q whose arc voltages sum to σ lifts to cycles of length $q \cdot m/\gcd(\sigma, m)$. Hence

Lemma 2. *If (H_{12}, φ) has girth ≥ 12 then $\sigma_C(\varphi) \not\equiv 0 \pmod{m}$ for every cycle C of H_{12} of length < 12, where σ_C is the sum of the arc voltages picked up by traversing C once, each arc counted $+1$ if traversed tail to head and -1 otherwise.*

The condition is *necessary only*, which is the safe direction for a nonexistence proof and the only direction used in Section 5: unsatisfiability of the necessary system rules out girth ≥ 12, hence a fortiori girth exactly 12. (Identifying the three pinned fibres can only merge vertices, which never lengthens a cycle; closed walks and lollipops can only lower the girth further.)

The census of cycles of H_{12} shorter than 12 is complete and finite: there are

$$18 + 27 + 82 = 127 \text{ cycles of lengths } 6, 8, 10, \quad \text{i.e. } \mathbf{254} \text{ directed,}$$

and there is no cycle of length below 6 and none of odd length (the graph is bipartite). The 254 is the source's own published integer for H_{12}, [1, p. 27]: "There are 254 such cycles". All 127 coefficient vectors have entries in $\{-1, 0, +1\}$, with arity histogram $\{1{:}16,\ 2{:}44,\ 3{:}4,\ 4{:}55,\ 6{:}6,\ 8{:}2\}$. The 16 arity-one constraints force 16 of the 20 arcs to be nonzero; the four arcs *not* so forced are exactly w, s, p, h, each of whose single-arc fundamental cycle has length 12.

4. The existence half: three witnesses

Theorem 3. *For each $m \in \{7, 8, 9\}$ the voltage assignment φ listed below makes (H_{12}, φ) a connected bipartite $(\{3, m\}; 12)$-graph on $41m + 3$ vertices, with $63m$ edges, no loop and no multiple edge, exactly three vertices of degree m and $41m$ of degree 3, and girth exactly 12.*

In the coordinate order $[a, b, c, d, e, j, q, u, w, s, l, g, f, k, r, v, \alpha, t, p, h]$:

W_7, a $(\{3, 7\}; 12)$-graph on **290** vertices:

```
phi = [1, 2, 2, 1, 2, 4, 6, 3, 0, 5, 5, 6, 1, 5, 3, 1, 1, 4, 0, 3]
```

A shortest cycle, as lift vertices (base vertex, fibre index):

```
(y11,5) (y1,5) (y,5) (y0,5) (y01,5) (y010,5)
(x0100,3) (z101,4) (z10,4) (z1,4) (z11,4) (x110,3)
```

W_8, a $(\{3, 8\}; 12)$-graph on **331** vertices:

```
phi = [1, 2, 2, 1, 3, 7, 3, 1, 0, 4, 1, 7, 1, 3, 7, 3, 1, 7, 0, 4]
(x100,7) (y10,0) (y1,0) (y,0) (y0,0) (y01,0)
(y011,0) (x1110,7) (x111,7) (x11,7) (x1,7) (x10,7)
```

W_9, a $(\{3, 9\}; 12)$-graph on **372** vertices — the order the authors name on their p. 28:

```
phi = [1, 2, 2, 1, 7, 6, 3, 1, 0, 5, 1, 3, 1, 3, 6, 7, 1, 3, 0, 5]
(x0,7) (x00,7) (z0,7) (z,7) (z1,7) (z10,7)
(z101,7) (x1110,7) (x111,7) (x11,7) (x1,7) (x,7)
```

Proof. Each witness is a finite exhibited object and every assertion of Theorem 3 is a finite computation on it, carried out on the $41m+3$ vertex lift built from Section 2: count the vertices and edges, check for loops and repeated pairs, take the degree histogram, run one breadth-first search for connectivity, and run breadth-first search from every vertex, capped at depth 6, for the girth. For a graph of girth $g \leq 12$ there is a root on a shortest cycle from which the two arcs of that cycle meet at depth $\lfloor g/2 \rfloor \leq 6$, so the cap loses nothing, and the minimum of $d(u) + d(w) + 1$ over the non-tree edges of all those searches is the girth exactly. The three displayed 12-cycles are the certificates that the girth is not larger: each is 12 distinct lift vertices whose 12 consecutive pairs are edges of the lift, which one checks directly against the tables of Section 2. That measurement is the only computation the existence half depends on, and it uses no solver. □

Witnesses keeping the published values of a, b, c, d. Theorem 3 is unrestricted. With a, b, c, d fixed at the source's Table-4 values $1, -1, -1, 1$ and only the remaining 16 arcs free, witnesses satisfying every assertion of Theorem 3 still exist at all three values of m:

```
m=7  phi = [1, 6, 6, 1, 6, 5, 2, 5, 0, 5, 3, 2, 1, 6, 4, 6, 5, 4, 2, 5]
m=8  phi = [1, 7, 7, 1, 6, 5, 5, 1, 0, 3, 6, 1, 1, 4, 7, 5, 1, 2, 0, 2]
m=9  phi = [1, 8, 8, 1, 1, 6, 3, 1, 0, 7, 1, 6, 2, 4, 5, 8, 1, 2, 0, 3]
```

(here $6 = -1$ in $\mathbb{Z}_7$, $7 = -1$ in $\mathbb{Z}_8$, $8 = -1$ in $\mathbb{Z}_9$). For $m \leq 6$ the restricted cell is empty a fortiori, since by Theorem 4 the full cell is.

5. The nonexistence half: $m = 3, 4, 5, 6$

Theorem 4. *For each $m \in \{3,4,5,6\}$ there is* no *$\varphi \in \mathbb{Z}_m^{20}$ for which (H_{12}, φ) has girth ≥ 12; in particular none for which it has girth exactly 12.*

Proof. By Lemma 2 it suffices to exhibit, for each m, a set of arcs and a set of short-cycle constraints supported on those arcs alone which is already unsatisfiable over $\mathbb{Z}_m$: a partial assignment violating a necessary constraint cannot be completed, so the whole cell $\mathbb{Z}_m^{20}$ is decided. The following four sets work; in each case *every* tuple over the listed arcs is enumerated flat, with no search and no propagation, and none survives.

m	arcs	tuples	survivors
3	g, f, k, p, h	$3^5 = 243$	0
4	u, w, s, r, v, α, t	$4^7 = 16{,}384$	0
5	$q, u, w, s, r, v, \alpha, t$	$5^8 = 390{,}625$	0
6	$q, u, w, s, r, v, \alpha, t$	$6^8 = 1{,}679{,}616$	0

The constraints used are, in each case, exactly those of the 127 of Section 3 whose support lies inside the listed arc set; they are recomputed from the tables of Section 2 by the accompanying program. No enumeration over $\mathbb{Z}_m^{20}$ itself is performed or needed: what is enumerated is only the four small cores in the table, and the orders $41m + 3$ that the four empty cells would have given are 126, 167, 208 and 249. $\square$

$m = 3$ is a pencil-and-paper proof. For $m = 3$ the ten constraints supported on g, f, k, p, h read, over $\mathbb{Z}_3$,

$$g \neq 0, \quad f \neq 0, \quad k \neq 0, \quad f \neq k, \quad p \neq f, \quad p \neq h,$$
$$h \neq g, \quad h \neq k, \quad h \neq g + k, \quad f - k - p + h \neq 0.$$

(For instance $f - k - p + h \neq 0$ comes from the octagon $x0100 \xrightarrow{f} z101 \xleftarrow{p} x1110 - x111 - x1111 \xrightarrow{h} z110 \xleftarrow{k} x0101 - x010 - x0100$.) Since g, f, k are nonzero they lie in $\{1, 2\}$, and $f \neq k$ forces $\{f, k\} = \{1, 2\}$. Four cases:

- $g = 1, k = 1$ (so $f = 2$): $h \neq 1$ and $h \neq g + k = 2$, so $h = 0$; then $p \neq f = 2$ and $p \neq h = 0$, so $p = 1$; but $f - k - p + h = 2 - 1 - 1 + 0 = 0$. Contradiction.
- $g = 1, k = 2$ (so $f = 1$): $h \neq 1$, $h \neq 2$, $h \neq g + k = 0$ — no value remains.
- $g = 2, k = 1$ (so $f = 2$): $h \neq 2$, $h \neq 1$, $h \neq g + k = 0$ — no value remains.
- $g = 2, k = 2$ (so $f = 1$): $h \neq 2$ and $h \neq g + k = 1$, so $h = 0$; then $p \neq 1$ and $p \neq 0$, so $p = 2$; but $f - k - p + h = 1 - 2 - 2 + 0 = -3 \equiv 0$. Contradiction.

So the $m = 3$ half needs no computer. At $m = 3$ the three pinned vertices have degree $m = 3$ as well, so the lift is *cubic*, on $41 \cdot 3 + 3 = 126$ vertices, and 126 is the order of the $(3, 12)$-cage. A cubic graph of girth 12 on 126 vertices *is* a $(3, 12)$-cage, and the $(3, 12)$-cage is unique [6]. Hence

Corollary 5. *The* $(3,12)$*-cage is not a* $\mathbb{Z}_3$ *lift of* H_{12}*; equivalently,* H_{12} *does not present the Tutte* 12*-cage* [5] *as a* $\mathbb{Z}_3$ *lift.*

Remark. Two riders on Corollary 5. (i) The *uniqueness* of the $(3,12)$-cage is quoted from the literature [6] and was not re-verified here; what is proved here is the emptiness of $\mathbb{Z}_3^{20}$, plus the arithmetic $41 \cdot 3 + 3 = 126$. (ii) The notation $n(\{3,3\};12)$ is not used anywhere in this note and should not be: the framework in which every other number here is stated requires the two degrees to be distinct, so the $m = 3$ cell lies outside it. Write "the $(3,12)$-cage", or "$(H_{12}, 3)$".

Remark (the boundary of Theorem 4). Theorem 4 is a statement about 20-arc voltage assignments on the graph of Section 2, in the switching normal form of Section 3. It does *not* say $n(\{3,m\};12) > 41m + 3$ for those m, and nothing here may be read that way. It is a nonexistence statement about one construction.

6. Consequences for the published bounds

The published record table for bi-regular cages is compiled by Goedgebeur, Jooken and Van den Eede [2]; the four rows relevant here read, in their version of record,

m	lower bound	upper bound
6	235	$250 \to 243$
7	272	334
8	308	374
9	344	$374 \to 360$

where "$x \to y$" is their own notation for "x from the literature, y our improvement". Theorem 3 therefore gives

Corollary 6. $n(\{3,7\};12) \leq 290$ *and* $n(\{3,8\};12) \leq 331$.

Both lie strictly inside the published feasible window — $\lceil 109 \cdot 7/3 \rceil + 17 = 272 < 290$ and $\lceil 109 \cdot 8/3 \rceil + 17 = 308 < 331$ — so the remaining gaps are 18 and 23.

Attribution. The 334 at $m = 7$ belongs to Araujo-Pardo, Balbuena, López-Chávez and Montejano [3], Theorem 2.2 — a reference of [1] itself, not a bound of [1]: the source's own Table 1 (p. 4) prints $\mathrm{rec}(\{3,7\};12) = 376$, i.e. it does not cite the better 334 that sits in its own bibliography. So no sentence here improves the source's authors at $m = 7$. The 374 at $m = 8$ *is* theirs (Corollary 2.4 of [1]); only there does this note improve a bound of [1].

Adjacent work, and what it does not settle. [2] is the current compilation of the $n(\{r,m\};g)$ bounds and the nearest work to this note: its Section 4 generalises the *gluing* theorem of [1] — "In [2] Aguilar, Araujo-Pardo, and Berman generalized Theorem 3 from [5]... we generalize their theorem a) for all girths $g \geq 5$, b) for more pairs (r, m), c) by improving their bound" — and not the H_{12} voltage-graph construction. It therefore says nothing about voltage assignments on H_{12} for $m < 10$, and its girth-12 rows leave the $m = 7$ and $m = 8$ upper bounds 334 and 374 with an empty improvement column; its own improvements in this range are at $m = 6$ ($250 \to 243$) and $m = 9$ ($374 \to 360$), the latter being the 360 that keeps the $m = 9$ witness below from carrying a record claim. [3], Theorem 2.2, supplies the 334 by a construction unrelated to H_{12} and settles nothing about it; [4], Lemma 3.4, supplies only the lower bound $\lceil 109m/3 \rceil + 17$ and no upper bound. None of the three addresses the sub-bullet's question, and none of them settles the identification of Section 7.

The $m = 8$ bound against Table 1 of the source. Table 1 of [1], p. 4, prints rec$(\{3, 8\}; 12) =$ **304**, which is *below* our 331, so a referee reading that page will read Corollary 6 as false; the claim should not be quoted without this paragraph. We read that 304 as a misprint for 374. No erratum is available to us and none is verified here; what is checked is the following. (i) $304 < 308$, and 308 is the lower bound printed in the adjacent column of the same row, reproduced by $\lceil 109 \cdot 8/3 \rceil + 17 = 291 + 17 = 308$: so the printed 304 is inconsistent with the lower bound of its own row. (ii) Table 1's technique column names Corollary 2.4 of [1] (p. 7) for that entry: "There exist $(3, m; 12)$-graphs of order: $41m + \frac{m}{3} + 2$ for $m = 3k$; $41m + \frac{m}{3} + 86 + \frac{2}{3}$ for $m = 3k+1$; $41m + \frac{m}{3} + 43 + \frac{1}{3}$ for $m = 3k+2$", and at $m = 8 = 3 \cdot 2 + 2$ that is $328 + \frac{8}{3} + 43 + \frac{1}{3} = 374$, while in exact rational arithmetic the same corollary reproduces every other rec$(\{3, m\}; 12)$ entry of Table 1 (252, 250, 250, 376, 374 at $m = 4, 5, 6, 7, 9$). (iii) [2] print 374 for that row, attributed to the same corollary, with an empty improvement column.

The $m = 9$ witness carries no record claim. $372 > 360$, and 360 is published [2]. The 372 of Theorem 3 realises the order the authors of [1] name on their p. 28, but it improves nothing, and no sentence here may say that it does. The orders 413 at $m = 10$ and 454 at $m = 11$ are Theorem 5.6 of [1] and not ours. And $m = 6$ carries nothing either: [2]'s improved $n(\{3, 6\}; 12) \leq 243$ is below the order 249 that $(H_{12}, 6)$ would have had — moot in any case, since Theorem 4 empties that cell.

7. Reading H_{12} off the source's tables

Tables 3 and 4 of [1], and the prose on its p. 27, cannot be transcribed literally into a graph whose 41 non-pinned vertices all have degree 3: they name leaves that the same page deletes. Section 2 therefore prints the graph we read from [1, Figure 16], and the two places where that reading departs from the printed tables are these.

(1) **Two endpoint misprints, against the source's own deletion list.** Tables 3 and 4 print the x-leaf sequence "… $x1010$, $x1001$ … " where the surviving leaf set forces $x1011$ ($x1001$ is itself one of the

eight deleted leaves, and is used twice while $x1011$ is never used); and Table 4's third row gives starting leaves $x1100, x1101$ for the arcs p, h, although both are deleted on p. 27, the surviving leaves still needing a z-arc being $x1110, x1111$.

(2) **The p. 27 prose "we added new labeled directed arcs from $x100$ to $y10$, $z10$ and from $x110$ to $y111$ and $z111$" cannot be read literally**, since $y111$ and $z111$ are both in the deletion list; we read it as Table 4's $x110 \to y11, z11$, which is what gives $y11, z11$ degree 3.

What supports the identification, and what does not. On the reading above all 41 non-pinned vertices are cubic, and the short-cycle census of Section 3 reproduces the source's own published 254 and its $41m + 3$; restoring the eight deleted leaves reproduces its $49m + 3$ and 441. Those are the whole of the evidence offered here. We do *not* supply an independent, arc-by-arc transcription of Figure 16, and the accompanying program does not check one: it is given the graph of Section 2 and checks statements about that graph. Nor do we claim that the reading above is the only reading with all 41 vertices cubic. Accordingly Proposition 1, Theorems 3 and 4 and Corollaries 5 and 6 are statements about the graph printed in Section 2, and this note does not claim more. In particular it does not claim to answer the question of [1]: that would need the printed graph to be the H_{12} of [1, Figure 16], which is exactly what is not established here and what a referee would have to check against the source, arc by arc. Corollary 6 is the one consequence unaffected by this, since its two witnesses are exhibited and checked as graphs in their own right.

8. What is not settled

- **The identification of the printed graph with** [1, Figure 16] **is not established** (Section 7), so the question of [1] is not claimed to be answered, in either direction.
- **The second question of the sub-bullet — the different graph H'_{12} — is completely untouched and remains open, at every m.** Nothing here rewires the leaf-joining arcs of T_{12}, builds a 16-cycle among the remaining leaves, or interlaces two 8-cycles differently, and nothing here says whether such an H'_{12} exists at any m.
- **The *second* sub-bullet of the same page is a different question and is not answered here.** Verbatim, [1, p. 29]: "Find voltage assignments to produce $(G_{12}; m)$ graphs with the construction given in Theorem 5.5 for any $4 \leq m \leq 8$, or show no such graphs exist". That is the G_{12} family of order $49m + 3$, not the $41m + 3$ family treated here.
- **Optimality is open.** Whether 290 and 331 are the true values of $n(\{3, 7\}; 12)$ and $n(\{3, 8\}; 12)$ is not addressed; the published lower bounds 272 and 308 stand untouched. No attempt was made to raise

them, and none to find a smaller $(\{3,7\};12)$ or $(\{3,8\};12)$ graph outside this family.

- **Counting is not claimed**, at any m: the number of assignments passing the necessary system of Lemma 2 is not the number giving girth 12.
- **Isomorphism questions are unexamined:** whether the witnesses of Section 4 are pairwise non-isomorphic, and whether any is isomorphic to a known graph, was not checked.
- **The uniqueness of the $(3,12)$-cage is an inherited external fact**, used only in Corollary 5 and flagged there.
- **Nonexistence is about this construction, not about the cage numbers** — see the remark closing Section 5. In particular the emptiness at $m = 4,5,6$ says nothing about [2]'s $n(\{3,4\};12) \leq 220$, $n(\{3,5\};12) \leq 230$, $n(\{3,6\};12) \leq 243$.

9. Verification

Every computational assertion of Sections 2–6 is re-derived, from the objects printed above and from nothing else, by the accompanying program `verify.py` (Python 3.9+, standard library only, exact integer and `Fraction` arithmetic, no floating point in any decision). It rebuilds the base graph from the tables of Section 2 and checks Proposition 1; recomputes the short-cycle census and the arity histogram of Section 3, reproducing the source's published 254; builds all six lifts and measures order, edge count, loops, repeated pairs, degree histogram, connectivity and exact girth, and checks the three displayed 12-cycles edge by edge; re-runs the four flat enumerations of Theorem 4 and the four-case argument for $m = 3$; and re-derives the arithmetic of Section 6, including Corollary 2.4 of [1] in exact rationals at $m = 4, \ldots, 9$. Its recorded run is in `verify.output.txt`. Its output also carries arithmetic side points this note does not state — among them the total size of the four cells emptied by Theorem 4, and evaluations of [2]'s Theorem 4.2 and of Table 1 of [1]; the Theorem 4.2 checks evaluate selected parameter choices and are not minimisations over its cases and parameters, and no check of the run verifies an erratum. The program is a convenience, not a dependency: the existence half reduces to one breadth-first search on a 290-vertex graph, and the $m = 3$ half is the four-case argument above.

What `verify.py` does not cover. It does not check that the graph of Section 2 is the graph of [1, Figure 16]: the program is handed that graph, and the readings of Section 7 are assertions about the source's tables, which are not reprinted here. A reader who wants them checked must fetch Tables 3 and 4 and Figure 16 from [1]. Beyond that, the program's own closing scope note lists the inherited external facts it does not re-run, among them the uniqueness of the $(3,12)$-cage used in Corollary 5 and the record values 334, 374 and 360, which are transcribed from the literature and not recomputed.

References

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